

A Counterexample to Convergence of the Nie–Tang KKT-Exclusion Algorithm*

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Abstract

Nie and Tang’s Algorithm 3.1 for Nash equilibrium problems of polynomials selects a KKT point by minimizing a positive definite quadratic, checks whether the selected point is a Nash equilibrium, and, if not, adds best-response inequalities that exclude the selected point. Their finite-termination theorem applies when the set of KKT points that are not Nash equilibria is finite, and they explicitly leave the convergence behavior in the infinite-spurious-KKT case unresolved. This note gives a two-player polynomial game, with a nonsingular constraint tuple and with pure Nash equilibria, for which the exact KKT-exclusion algorithm generates infinitely many non-equilibrium KKT points converging to another non-equilibrium KKT point. The positive definite selector can be chosen from a nonempty Euclidean open set of selectors. Consequently, under the standard interpretation of “generic” as membership in a prescribed Zariski-open dense set, the selector can also be chosen generic. In that standard sense the construction is a full negative solution to the convergence question; without formalizing the word “generic,” the explicit positive-definite-selector theorem should be read as rigorous partial progress.

1 Status and relation to the Nie–Tang problem

Nie and Tang formulate, for a polynomial Nash equilibrium problem (NEPP), a polynomial optimization problem over the KKT set and then iteratively add inequalities of the form

$$f_i(v_i, x_{-i}) - f_i(x_i, x_{-i}) \geq 0,$$

where v_i is a best response to the current candidate. In their Algorithm 3.1, the KKT point is selected by minimizing a positive definite quadratic

$$q_\Theta(x) := [x]_1^T \Theta [x]_1, \quad [x]_1 = (1, x_1, \dots, x_n)^T,$$

with Θ chosen generic and positive definite. They prove finite termination when the set $G \setminus G^*$ of KKT points that are not Nash equilibria is finite. The remaining case, where $G \setminus G^*$ is infinite, is the issue considered here.

The result below is stronger than merely producing infinitely many non-equilibrium KKT points. The game has Nash equilibria, but the exact algorithm repeatedly selects non-equilibrium KKT points

$$(0, 0, b), (0, 0, \rho b), (0, 0, \rho^2 b), \dots, \quad 0 < \rho < 1,$$

*This note is a cleaned-up verification of a proposed construction for the open convergence issue left after Theorem 3.2 of Nie and Tang.

and the limit $(0, 0, 0)$ is itself not a Nash equilibrium. Thus the failure is not just non-finite termination; it is convergence to the wrong point.

The only interpretive point is the word “generic” in the selector. Section 6 proves that the phenomenon holds for every selector in a nonempty Euclidean open subset of the positive definite cone. Hence this open set intersects every prescribed Zariski-open dense generic set of positive definite selectors. With the usual algebraic-geometric meaning of generic, this gives a genuine counterexample to Algorithm 3.1 as stated. If one refuses to attach this standard meaning to the word because the paper does not name the exact exceptional algebraic set, then Theorem 1 remains a rigorous positive-definite-selector counterexample and Section 6 explains the remaining genericity patch.

2 The polynomial game

Define

$$P(t) := (t - 1)^2 - \frac{1}{4}, \quad p(t) := t^2 P(t), \quad a := \frac{3 + \sqrt{3}}{4}.$$

Then

$$p'(t) = 2t \left(2t^2 - 3t + \frac{3}{4} \right).$$

The critical points are 0, $(3 - \sqrt{3})/4$, and $a = (3 + \sqrt{3})/4$. From the sign pattern of p' and the fact that $p(t) \rightarrow +\infty$ as $|t| \rightarrow \infty$, the point a is the unique global minimizer of p , and

$$p(a) = -\frac{9}{64} - \frac{3\sqrt{3}}{32} < 0.$$

Set

$$\rho := \frac{1}{2a} = \frac{2}{3 + \sqrt{3}} \in (0, 1).$$

Consider the following two-player game. Player 1 chooses $(x, z) \in \mathbb{R}^2$ and has no constraints. Player 2 chooses $y \in [0, b]$, where $b > 0$ will be chosen sufficiently small. The objectives are

$$f_1(x, z, y) := x^2 P(xy) + a^2 z^2 + (xy - a)^2 (z^4 - 2z^2), \tag{1}$$

$$f_2(y, x, z) := (x^2 + z^2)y^2. \tag{2}$$

Player 2's feasible interval is represented by

$$g_{2,1}(y) = y \geq 0, \quad g_{2,2}(y) = b - y \geq 0.$$

The constraint tuple for Player 2 is nonsingular in the Nie–Tang sense. Indeed, the associated matrix is

$$G_2(y) = \begin{bmatrix} 1 & -1 \\ y & 0 \\ 0 & b - y \end{bmatrix},$$

and its two columns are linearly independent for every $y \in \mathbb{C}$, since y and $b - y$ cannot both vanish when $b \neq 0$. Player 1's empty constraint tuple is nonsingular by convention.

3 A uniform best-response lemma

Lemma 1. *There exists $\bar{b} > 0$ such that, for every $0 < y \leq \bar{b}$, Player 1's problem*

$$\min_{(x,z) \in \mathbb{R}^2} f_1(x, z, y)$$

has the unique global minimizer

$$(x, z) = \left(\frac{a}{y}, 0 \right).$$

Proof. Put $t = xy$. For $y > 0$,

$$f_1(x, z, y) = \frac{p(t)}{y^2} + a^2 z^2 + (t - a)^2 (z^4 - 2z^2). \quad (3)$$

Because p has a unique global minimizer at a and $p''(a) > 0$, the function

$$t \mapsto \frac{p(t) - p(a)}{(t - a)^2}$$

when extended continuously at $t = a$ by $p''(a)/2$, has a positive infimum. Denote this infimum by $m > 0$. Also,

$$z^4 - 2z^2 = (z^2 - 1)^2 - 1 \geq -1.$$

Subtracting $p(a)/y^2$ from (3) gives

$$\begin{aligned} f_1(x, z, y) - \frac{p(a)}{y^2} &\geq \frac{p(t) - p(a)}{y^2} + a^2 z^2 - (t - a)^2 \\ &\geq \left(\frac{m}{y^2} - 1 \right) (t - a)^2 + a^2 z^2. \end{aligned}$$

Choose any $\bar{b} < \sqrt{m}$. If $0 < y \leq \bar{b}$, the right-hand side is nonnegative and vanishes only when $t = a$ and $z = 0$. Since $t = xy$, the unique global minimizer is $(x, z) = (a/y, 0)$. $\square$

From now on fix any

$$0 < b \leq \bar{b}.$$

4 The KKT set and the Nash equilibria

We use the standard KKT sign convention for inequalities $g \geq 0$ in a minimization problem: stationarity is

$$\nabla f - \sum_j \lambda_j \nabla g_j = 0, \quad \lambda_j \geq 0, \quad \lambda_j g_j = 0.$$

Equivalently, $\sum_j \lambda_j \nabla g_j = \nabla f$.

Proposition 1. *The real KKT set of the game is*

$$\mathcal{K} = \{(0, 0, y) : 0 \leq y \leq b\} \cup \left\{ \left(0, \frac{1}{\sqrt{2}}, 0 \right), \left(0, -\frac{1}{\sqrt{2}}, 0 \right) \right\}.$$

Moreover, every point $(0, 0, y)$ with $0 \leq y \leq b$ is a KKT point.

Proof. First, let $(x, z) = (0, 0)$. The gradients of f_1 with respect to Player 1's variables vanish:

$$\frac{\partial f_1}{\partial x}(0, 0, y) = 0, \quad \frac{\partial f_1}{\partial z}(0, 0, y) = 0.$$

Also $\partial f_2 / \partial y = 2(x^2 + z^2)y = 0$. Taking both Player 2 multipliers to be zero gives stationarity and complementarity. Hence $(0, 0, y)$ is KKT for every $y \in [0, b]$.

Conversely, suppose (x, z, y) is a KKT point. If $0 < y < b$, both constraints of Player 2 are inactive, so stationarity for Player 2 gives

$$2(x^2 + z^2)y = 0,$$

and hence $(x, z) = (0, 0)$. If $y = b$, then the lower-bound constraint $g_{2,1} = y$ is inactive, so its multiplier is zero. The upper-bound constraint has gradient -1 , and stationarity gives

$$-\lambda_{2,2} = 2(x^2 + z^2)b.$$

Since $\lambda_{2,2} \geq 0$ and $b > 0$, this is possible only when $(x, z) = (0, 0)$. Thus every KKT point with $y > 0$ lies on the line segment already found.

It remains to treat $y = 0$. The upper-bound constraint is inactive, so $\lambda_{2,2} = 0$, and Player 2 stationarity gives $\lambda_{2,1} = 0$. Thus only Player 1 stationarity matters. At $y = 0$,

$$f_1(x, z, 0) = \frac{3}{4}x^2 + a^2(z^4 - z^2),$$

so

$$\frac{\partial f_1}{\partial x}(x, z, 0) = \frac{3}{2}x, \quad \frac{\partial f_1}{\partial z}(x, z, 0) = 2a^2z(2z^2 - 1).$$

Therefore $x = 0$ and $z \in \{0, \pm 1/\sqrt{2}\}$. This gives exactly the displayed KKT set. $\square$

Proposition 2. *The game has Nash equilibria, namely*

$$\left(0, \frac{1}{\sqrt{2}}, 0\right) \quad \text{and} \quad \left(0, -\frac{1}{\sqrt{2}}, 0\right).$$

The points $(0, 0, y)$ with $0 < y \leq b$ are not Nash equilibria, and $(0, 0, 0)$ is not a Nash equilibrium.

Proof. At $y = 0$,

$$f_1(x, z, 0) = \frac{3}{4}x^2 + a^2(z^4 - z^2).$$

The term $3x^2/4$ is minimized uniquely at $x = 0$, and $z^4 - z^2$ is minimized at $z = \pm 1/\sqrt{2}$. Hence Player 1's global minimizers against $y = 0$ are exactly $(0, \pm 1/\sqrt{2})$. Against $(x, z) = (0, \pm 1/\sqrt{2})$, Player 2 minimizes

$$f_2(y, 0, \pm 1/\sqrt{2}) = \frac{1}{2}y^2$$

over $[0, b]$, so the unique response is $y = 0$. The two displayed points are therefore Nash equilibria.

If $0 < y \leq b$, Lemma 1 gives the unique Player 1 best response $(a/y, 0)$. Since

$$f_1(0, 0, y) = 0, \quad f_1\left(\frac{a}{y}, 0, y\right) = \frac{p(a)}{y^2} < 0,$$

the point $(0, 0, y)$ is not a Nash equilibrium. Finally,

$$f_1(0, 0, 0) = 0, \quad f_1\left(0, \frac{1}{\sqrt{2}}, 0\right) = -\frac{a^2}{4} < 0,$$

so $(0, 0, 0)$ is also not a Nash equilibrium. $\square$

Thus $\mathcal{K} \setminus \mathcal{K}^*$ is infinite; it contains the whole punctured segment $\{(0, 0, y) : 0 < y \leq b\}$ and also the endpoint $(0, 0, 0)$.

5 The exclusion cuts and the generated sequence

Let Θ be positive definite and define

$$q_{\Theta}(x, z, y) := [1, x, z, y]^T \Theta [1, x, z, y].$$

For the first theorem in this section, we assume that Θ satisfies the following two selector inequalities:

(S1) The derivative of the one-variable restriction is strictly negative on $[0, b]$:

$$\frac{d}{dy} q_{\Theta}(0, 0, y) < 0 \quad (0 \leq y \leq b).$$

(S2) The endpoint $(0, 0, 0)$ is strictly cheaper than the two equilibrium KKT points at $y = 0$:

$$q_{\Theta}(0, 0, 0) < q_{\Theta}\left(0, \frac{1}{\sqrt{2}}, 0\right), \quad q_{\Theta}(0, 0, 0) < q_{\Theta}\left(0, -\frac{1}{\sqrt{2}}, 0\right).$$

Define

$$s_0 := b, \quad s_{\ell+1} := \rho s_{\ell}, \quad \rho = \frac{1}{2a}.$$

Equivalently, $s_{\ell} = \rho^{\ell} b$.

Lemma 2. *Suppose Algorithm 3.1 has selected the KKT point $(0, 0, s)$ with $0 < s \leq b$. Then Player 1's best response is uniquely $(a/s, 0)$, and the cut added for Player 1 restricts the KKT line segment $\{(0, 0, y) : 0 \leq y \leq s\}$ to*

$$\left\{ (0, 0, y) : 0 \leq y \leq \frac{s}{2a} \right\}.$$

The three KKT points with $y = 0$ all satisfy the cut.

Proof. The best response statement is Lemma 1. The corresponding cut is

$$f_1\left(\frac{a}{s}, 0, y\right) - f_1(x, z, y) \geq 0. \tag{4}$$

On the KKT line $(x, z) = (0, 0)$, the second term is zero and

$$f_1\left(\frac{a}{s}, 0, y\right) = \frac{a^2}{s^2} P\left(\frac{ay}{s}\right).$$

Hence (4) is equivalent on the KKT line to

$$P\left(\frac{ay}{s}\right) \geq 0.$$

Since

$$P(t) = \left(t - \frac{1}{2}\right) \left(t - \frac{3}{2}\right),$$

we have $P(t) \geq 0$ exactly when $t \leq 1/2$ or $t \geq 3/2$. On $0 \leq y \leq s$,

$$0 \leq \frac{ay}{s} \leq a < \frac{3}{2},$$

so only the first alternative can occur. Therefore the surviving part of the old line segment is precisely

$$0 \leq \frac{ay}{s} \leq \frac{1}{2}, \quad \text{i.e.} \quad 0 \leq y \leq \frac{s}{2a}.$$

At the $y = 0$ KKT points,

$$f_1\left(\frac{a}{s}, 0, 0\right) = \frac{a^2}{s^2}P(0) = \frac{3a^2}{4s^2} > 0,$$

while $f_1(0, 0, 0) = 0$ and $f_1(0, \pm 1/\sqrt{2}, 0) = -a^2/4$. Hence all three $y = 0$ KKT points satisfy the cut. $\square$

Theorem 1. *Let Θ be positive definite and satisfy (S1)–(S2). For the game (1)–(2), with exact solution of all subproblems, Algorithm 3.1 produces the infinite sequence*

$$u^{(\ell)} = (0, 0, s_\ell) = (0, 0, \rho^\ell b), \quad \ell = 0, 1, 2, \dots$$

Consequently,

$$u^{(\ell)} \longrightarrow (0, 0, 0),$$

but the limit is not a Nash equilibrium.

Proof. Let $\mathcal{F}_\ell$ denote the feasible KKT set in problem (3.10) of Algorithm 3.1 after ℓ rounds of cuts, with $\mathcal{F}_0 = \mathcal{K}$. We prove by induction that

$$\mathcal{F}_\ell = \{(0, 0, y) : 0 \leq y \leq s_\ell\} \cup \left\{ \left(0, \frac{1}{\sqrt{2}}, 0\right), \left(0, -\frac{1}{\sqrt{2}}, 0\right) \right\}. \quad (5)$$

For $\ell = 0$, this is exactly Proposition 1, since $s_0 = b$.

Assume (5) holds. Because $q_\Theta(0, 0, y)$ is strictly decreasing on $[0, b]$ by (S1), it is uniquely minimized on $\{(0, 0, y) : 0 \leq y \leq s_\ell\}$ at $y = s_\ell$. Also, since $s_\ell > 0$,

$$q_\Theta(0, 0, s_\ell) < q_\Theta(0, 0, 0),$$

and by (S2) the latter is strictly smaller than the selector values at $(0, \pm 1/\sqrt{2}, 0)$. Therefore the unique minimizer of q_Θ on $\mathcal{F}_\ell$ is

$$u^{(\ell)} = (0, 0, s_\ell).$$

By Proposition 2, this point is not a Nash equilibrium. Player 1's best response is uniquely $(a/s_\ell, 0)$ and yields negative improvement, while Player 2's objective is identically zero when $(x, z) = (0, 0)$, so Player 2 has no negative improvement. Thus Algorithm 3.1 adds exactly the Player 1 cut associated with $(a/s_\ell, 0)$. Lemma 2 says that the line segment in (5) is cut down to

$$\left\{ (0, 0, y) : 0 \leq y \leq \frac{s_\ell}{2a} \right\} = \{(0, 0, y) : 0 \leq y \leq s_{\ell+1}\},$$

and that the two additional $y = 0$ KKT points survive. Hence (5) holds with $\ell + 1$ in place of ℓ .

This proves the asserted iterate formula. Since $0 < \rho < 1$, the sequence converges to $(0, 0, 0)$. Proposition 2 shows that this limit is not a Nash equilibrium. $\square$

6 The selector can be chosen generic

It remains to justify that the failure is not an artifact of a specially tuned nongeneric selector. Let $\mathbb{S}^4$ be the vector space of real symmetric 4×4 matrices and let $\mathbb{S}_{++}^4$ be its positive definite cone. Define

$$\mathcal{U} := \{\Theta \in \mathbb{S}_{++}^4 : \Theta \text{ satisfies (S1) and (S2)}\}.$$

Lemma 3. *The set $\mathcal{U}$ is a nonempty Euclidean open subset of $\mathbb{S}_{++}^4$.*

Proof. Openness follows from strict inequalities on a compact interval and at two fixed points. More explicitly, the derivative of $y \mapsto q_\Theta(0, 0, y)$ depends continuously on (Θ, y) , so the strict negativity required in (S1) on the compact interval $[0, b]$ persists under sufficiently small perturbations of Θ . The two inequalities in (S2) are also strict finite-dimensional inequalities and therefore open.

To prove nonemptiness, choose any $c > b$ and $C > 0$, and set

$$Q_0(x, z, y) := C + x^2 + z^2 + (y - c)^2.$$

This is represented as $[1, x, z, y]^T \Theta_0 [1, x, z, y]$ with

$$\Theta_0 = \begin{bmatrix} C + c^2 & 0 & 0 & -c \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -c & 0 & 0 & 1 \end{bmatrix}.$$

The $(1, y)$ -block has determinant $C > 0$, and the remaining diagonal entries are positive, so Θ_0 is positive definite. Moreover,

$$Q_0(0, 0, y) = C + (y - c)^2$$

has derivative $2(y - c) < 0$ on $[0, b]$, since $c > b$, and

$$Q_0\left(0, \pm \frac{1}{\sqrt{2}}, 0\right) = Q_0(0, 0, 0) + \frac{1}{2} > Q_0(0, 0, 0).$$

Thus $\Theta_0 \in \mathcal{U}$. □

Theorem 2. *Let $\Gamma \subset \mathbb{S}^4$ be any nonempty Zariski-open dense set of symmetric matrices. Then*

$$\Gamma \cap \mathcal{U} \neq \emptyset.$$

Consequently, if the phrase “choose a generic positive definite matrix” in Algorithm 3.1 means that Θ must belong to some prescribed nonempty Zariski-open dense set, then Θ may be chosen generic and positive definite while still producing the nonconvergent sequence of Theorem 1.

Proof. The complement of a proper algebraic subset of $\mathbb{S}^4$ is Euclidean dense. Since Γ is Zariski-open dense, it is Euclidean dense. By Lemma 3, $\mathcal{U}$ is a nonempty Euclidean open set. Therefore $\Gamma \cap \mathcal{U}$ is nonempty. Any Θ in this intersection is positive definite, satisfies (S1)–(S2), and is generic in the prescribed Zariski-open sense. The conclusion then follows from Theorem 1. □

Corollary 1. *Under the standard algebraic-geometric interpretation of genericity, Algorithm 3.1 is not guaranteed to converge to a Nash equilibrium when the set of KKT points that are not Nash equilibria is infinite, even if a pure Nash equilibrium exists and all subproblems are solved exactly.*

Remark 1 (What is proved, and what is merely interpretive). The explicit selector $Q_0 = C + x^2 + z^2 + (y - c)^2$ already gives a rigorous counterexample for a positive definite selector. Theorem 2 upgrades this to a generic-selector counterexample whenever “generic” is understood in the usual Zariski-open dense sense. If the original algorithm is read as requiring membership in some unspecified set not known to be Zariski-open dense, then the completely unconditional statement is Theorem 1; in that deliberately conservative reading, the result is partial progress rather than a formally closed solution.

7 Conclusion

The game constructed above satisfies the structural assumptions needed to formulate the Nie–Tang KKT optimization: its objectives are polynomials, Player 2’s constraint tuple is nonsingular, and Nash equilibria exist. Nevertheless, the KKT set contains a continuum of non-equilibrium KKT points. The best-response cut generated at $(0, 0, s)$ removes the upper part of the current KKT segment but leaves the smaller segment $0 \leq y \leq s/(2a)$. A whole open set of positive definite quadratic selectors then chooses the new right endpoint. The exact iterates therefore form the geometric sequence

$$(0, 0, b), (0, 0, \rho b), (0, 0, \rho^2 b), \dots, \quad \rho = \frac{1}{2a} \in (0, 1),$$

which converges to $(0, 0, 0)$, a KKT point that is not a Nash equilibrium. Thus the infinite-spurious-KKT regime admits genuine nonconvergence of Algorithm 3.1 to Nash equilibrium.

References

- [1] Jiawang Nie and Xindong Tang, *Nash Equilibrium Problems of Polynomials*, Mathematics of Operations Research, 49(2):1065–1090, 2024. DOI: 10.1287/moor.2022.0334. Preprint: arXiv:2006.09490.